

Supporting information

Structural limits of single-barrier reform in algorithmic recourse: a formal series-system model with implications for digital health

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Components of this appendix are labelled alphabetically and should be cited as, for example, “Table A in S1 Appendix” and “Fig A in S1 Appendix”.

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Text A. Scope of claims

Throughout, numerical results (baseline 0.0018%, maximum single-barrier gain 0.0054%, three-way share 87.6%, and all Shapley/Sobol/Morris/bootstrap outputs) are **level-2** model illustrations generated from the chosen topology and parameterization; they are consistent with, but not independent confirmation of, Propositions 1–2 (level 1). Pass probabilities (Table A) are **level-3** provisional calibration assumptions. Statements about real recourse behavior or policy are **level-4** hypotheses.

Table A. Parameter derivation audit trail (all 11 barriers)

Table A. Parameter derivation audit trail. For each barrier: target construct, source domain, exact source statistic, mapping rule, resulting pass probability, plausibility range, and transportability limitation. Auditable from this package alone; mirrored in `data/parameter_sources/parameter_derivation.md`. Source locators identify the specific discussion or finding within each source rather than an exact page, table, or figure number; they were not confirmed to page-level precision against the primary sources.

Barrier (layer)	Source domain	Exact statistic [source]	<i>p</i>	Range	Mapping rule	Transportability limitation
Rapid Data Transmission (L1)	Credit reporting	Furnishers transmit adverse data to CRAs within 1–2 days [CFPB 2022, verify]	0.30	0.20–0.40	~30% intervene before propagation	Credit-furnishing timing; not measured for clinical/EHR data flows
Multi-System Integration (L1)	Credit reporting	20% had error on ≥1 of 3 reports [FTC 2013, verify]	0.55	0.40–0.65	Complement of cross-system error rate	Cross-bureau error; not clinical multi-system integration
Permanent Storage (L1)	Credit reporting	7-yr retention; <20% adverse tradelines removed in 4 yr [CFPB 2022; FCRA §605, verify]	0.45	0.35–0.55	~45% encounter unexpired adverse data	FCRA retention regime; not healthcare record retention
Error Detection Difficulty (L2)	Credit reporting	26% identified material errors [FTC 2013, verify]	0.35	0.25–0.45	~35% identify specific errors	Consumer credit review; not clinical error detection
Correction Process Barriers (L2)	Credit reporting	37% of disputes fully resolved [FTC 2013; CFPB 2022, verify]	0.35	0.25–0.45	~35% achieve full correction	Credit dispute process; not clinical data correction
Incomplete Correction Propagation (L2)	Credit reporting	Corrections do not auto-propagate across CRAs [CFPB 2022; FTC 2015, verify]	0.40	0.30–0.50	~40% propagate sufficiently	Inter-CRA propagation; not clinical systems

Barrier (layer)	Source domain	Exact statistic [source]	p	Range	Mapping rule	Transportability limitation
Awareness Gap (L3)	Civil legal	Help sought for ~1 in 4 substantial civil legal problems [LSC 2022, verify]	0.30	0.20–0.40	~30% become aware of recourse	Civil legal need; not algorithmic-decision awareness specifically
Record Access Barriers (L3)	Credit reporting	Statutory free-report access with practical barriers [FCRA §612; CFPB 2022, verify]	0.55	0.45–0.65	Most navigable barrier (~55%)	FCRA access; not clinical record access
Legal Knowledge Gap (L3)	Civil legal	39% believe they can use the legal system to protect their rights [LSC 2022, verify]	0.25	0.15–0.35	~25% know specific algorithmic rights	General legal confidence; not algorithm-specific rights knowledge
Legal Resource Barriers (L3)	Civil legal	46% cite cost; ~50% of applicants turned away [LSC 2022, verify]	0.40	0.30–0.50	~40% with knowledge access resources	Civil legal aid capacity; not algorithm-specific representation
Systemic Bias in Algorithms (L3)	Healthcare audit	Cost-for-need proxy bias; correction raises Black enrollment 17.7%→46.5% [Obermeyer 2019]	0.30	0.20–0.40	~30% recourse not defeated by residual system bias	Single commercial tool / one health system; NOT a population-wide bias rate (prior “62%” figure was erroneous and is corrected)

Table B. Individual barrier removal effects

Table B. Individual barrier removal effects. Baseline $P = 0.0018\%$; Δ = marginal effect of removing each barrier (deterministic; $\Delta_j = P(1/p_j - 1)$, Proposition 1). All $\Delta < 0.02\%$; the maximum is Legal Knowledge Gap ($\Delta = 0.0054\%$). Bootstrap 95% CIs ($n = 1,000$, seed 42) are provided in the repository (`individual_barrier_effects.csv`).

Barrier	Layer	P after removal (%)	Δ (%)
Rapid Data Transmission	Data Integration	0.0060	0.0042
Multi-System Integration	Data Integration	0.0033	0.0015
Permanent Storage	Data Integration	0.0040	0.0022
Error Detection Difficulty	Data Accuracy	0.0051	0.0033
Correction Process Barriers	Data Accuracy	0.0051	0.0033
Incomplete Correction Propagation	Data Accuracy	0.0045	0.0027
Awareness Gap	Institutional	0.0060	0.0042
Record Access Barriers	Institutional	0.0033	0.0015
Legal Knowledge Gap	Institutional	0.0072	0.0054
Legal Resource Barriers	Institutional	0.0045	0.0027
Systemic Bias in Algorithms	Institutional	0.0060	0.0042

Table C. Layer interaction decomposition

Table C. Baseline-relative factorial decomposition of achievable improvement. Values match the code, Fig 3 heatmap, and S-figures. (Corrects a stale main-text table in the prior submission.)

Effect	Share of achievable improvement (%)
L1 (Data Integration)	0.02
L2 (Data Accuracy)	0.03
L3 (Institutional)	0.36
L1 $\times$ L2	0.44
L1 $\times$ L3	4.51
L2 $\times$ L3	7.03
L1 $\times$ L2 $\times$ L3 (three-way)	87.60
Total	100.0

Table D. Shapley value attribution (seeded, seed 42). Fair allocation of achievable improvement across barriers, accounting for all removal orderings. Contributions are comparatively even (no barrier >11%) and span all three layers, consistent with interaction-driven structure.

Rank	Barrier	Layer	Shapley (%)
1	Legal Knowledge Gap	Institutional	10.95
2	Rapid Data Transmission	Data Integration	10.72
3	Systemic Bias in Algorithms	Institutional	10.53
4	Incomplete Correction Propagation	Data Accuracy	10.10
5	Awareness Gap	Institutional	9.67
6	Error Detection Difficulty	Data Accuracy	9.53
7	Correction Process Barriers	Data Accuracy	9.21
8	Legal Resource Barriers	Institutional	8.50
9	Permanent Storage	Data Integration	7.79
10	Record Access Barriers	Institutional	6.79
11	Multi-System Integration	Data Integration	6.21

Table E. Sobol first-order (S_1) and total-order (S_T) indices (seeded, seed 42). S_T exceeded S_1 for *all* barriers, but the $S_T - S_1$ interaction gaps are small and, for several barriers, within Monte-Carlo error at the current Saltelli sample size—so “substantially exceeded for all barriers” is not supported. Under seeding the smallest gaps fall on Correction Process/Legal Resource, underscoring that per-barrier orderings are sampling-sensitive (finalization option: increase n).

Barrier	S_1	S_T	$S_T - S_1$
Rapid Data Transmission	0.063	0.101	0.038
Multi-System Integration	0.073	0.103	0.030
Permanent Storage	0.059	0.097	0.038
Error Detection Difficulty	0.069	0.098	0.029
Correction Process Barriers	0.088	0.103	0.015
Incomplete Correction Propagation	0.074	0.101	0.027
Awareness Gap	0.075	0.100	0.025
Record Access Barriers	0.075	0.108	0.033
Legal Knowledge Gap	0.084	0.103	0.019
Legal Resource Barriers	0.094	0.106	0.011
Systemic Bias in Algorithms	0.065	0.098	0.034

Table F. Stability of key findings within the prespecified $\pm 10\%$ perturbation scenario ($n = 1,000$ resamples). Reported as stability within bounded scenarios, not robustness under arbitrary uncertainty; see S7 for coefficient-of-variation escalation.

Finding	Threshold	% resamples supporting
Three-way share high	>70%	100.0
Max individual effect small	<1%	100.0
Barriers needed for 90% success	≥ 10 of 11	100.0

Table D. Shapley value attribution

Table E. Sobol sensitivity indices

Table F. Robustness under bounded perturbation

Table G. Signal-to-noise and output variability

Table H. Robustness to alternative topologies

Table G. SNR and coefficient of variation (CV) by parameter-noise level (seeded, seed 42). Robustness is bounded: CV rises steeply above $\sim 15\%$ noise, reaching 109.8% at 25% and 128.7% at 30%. Robustness is therefore claimed only within bounded scenarios.

Noise (%)	SNR (dB)	Quality	CV (%)
1	14.8	Good	3.3
2	11.7	Good	6.8
5	7.8	Marginal	16.6
10	4.6	Marginal	35.1
15	2.8	Marginal	53.0
20	1.4	Marginal	73.0
25	-0.4	Poor	109.8
30	-1.1	Poor	128.7

Seeded canonical values (seed 42). These replace the prior unseeded values; the output is small and right-skewed, so CV point estimates are sampling-sensitive at the current n (finalization option: increase n to stabilize).

Table H. Correlated-barrier (Gaussian copula) and repeated-attempt robustness (seed 42, $n = 100,000$). The comparative conclusion (single-layer $\ll$ coordinated) is robust to correlation; the baseline value is correlation-sensitive; repeated attempts attenuate interaction dominance, delimiting the domain of validity.

Scenario	Baseline P	3-way share	max 1-barrier	max 1-layer
Independence ($\rho = 0$)	0.0018%	87.6%	0.0054%	0.36%
Copula $\rho = 0.1$	0.049%	84.5%	0.071%	0.97%
Copula $\rho = 0.2$	0.288%	81.9%	0.185%	1.85%
Copula $\rho = 0.3$	0.835%	79.9%	0.376%	2.74%
Copula $\rho = 0.4$	1.778%	78.5%	0.606%	3.71%
Copula $\rho = 0.5$	3.200%	77.4%	0.821%	4.50%
Block copula $\rho = 0.5$	0.151%	66.0%	0.142%	—
Repeated attempts $m = 2$	0.352%	53.3%	0.452%	5.70%
Repeated attempts $m = 3$	3.651%	25.2%	2.665%	16.88%

correlation sweep; under repeated attempts it attenuates, bounding the model’s validity to approximately single-pass serial recourse.

Supplementary figures

Code and data availability

All code, data, and a one-command `make` reproduction pipeline (seed 42) are at <https://github.com/Nyx-Dynamics/algorithmic-bias-epidemiology-academic>, archived at Zenodo (DOI:10.5281/zenodo.22312977). Key scripts: `barrier_visualization.py`, `sensitivity_analysis.py`, `copula_robustness.py`, `build_derivation_table.py`. Environment: Python 3.10+, NumPy, SciPy, Matplotlib. Random seed 42 (now enforced in all stochastic routines).

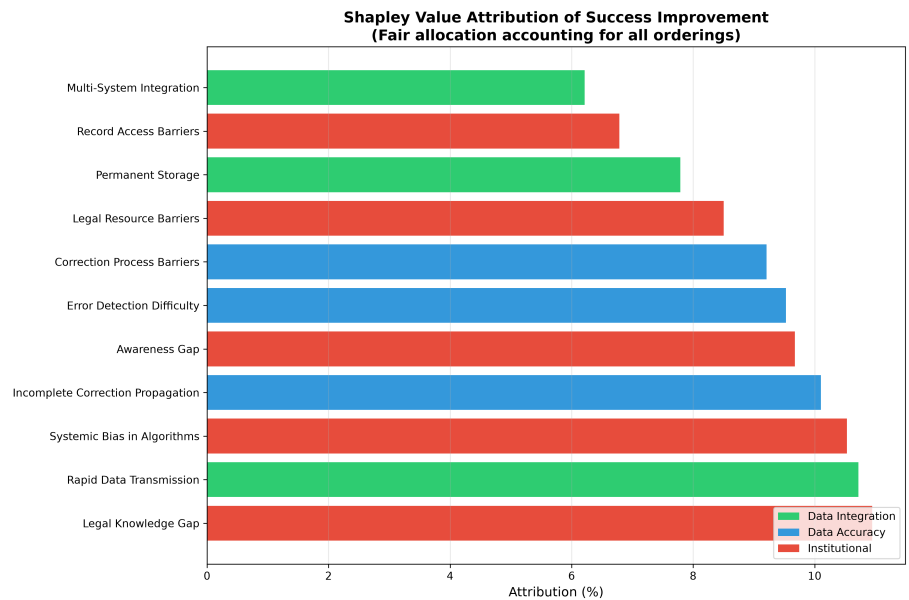


Fig A. Shapley value attribution (seeded). Fair allocation of achievable improvement across barriers, accounting for all removal orderings; the five highest contributors span all three layers, consistent with interaction-driven structure.

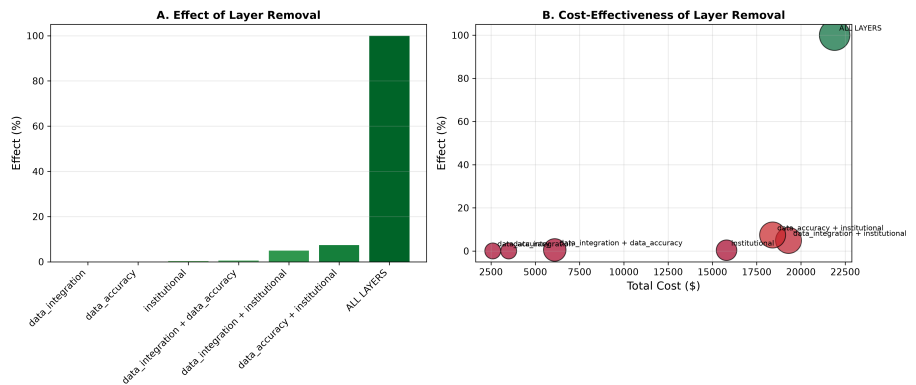


Fig B. Layer-removal effects. Single-layer removal yields negligible improvement; the largest two-layer combination is Data Accuracy + Institutional (~7.4%); only removal across all three layers reaches 100%.

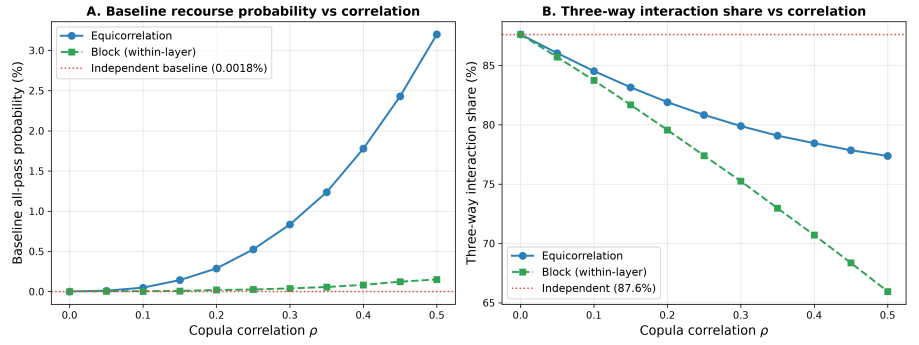


Fig C. Alternative-topology robustness. Baseline success P and three-way interaction share vs. Gaussian-copula correlation ρ , with repeated-attempt points ($m = 2, 3$). The comparative conclusion is robust to correlation; the baseline value is correlation-sensitive; repeated attempts attenuate interaction dominance.